

1. ALGORITHM

1. Start

2. Display the menu options (1 to 5) to select the pattern type.

3. Read the user's choice (choice).

4. If choice is between 1 and 5, prompt for the number of rows (n).

5. Execute based on choice:

- **Choice 1:** Loop i from 1 to n. Inner loop j from 1 to i: print *. Print newline.
- **Choice 2:** Loop i from 1 to n. Inner loop j from 1 to i: print i. Print newline.
- **Choice 3:** Loop i from 1 to n. Inner loop j from 1 to i: print j. Print newline.
- **Choice 4:** Set count = 1. Loop i from 1 to n. Inner loop j from 1 to i: print count, increment count. Print newline.
- **Choice 5:** Loop i from 1 to n. Inner loop j from 1 to i: print character 'A' + j - 1. Print newline.

6. If choice is outside 1–5, display "Invalid choice!".

7. Stop

2. TEST-CASE TABLE

Test Case	Pattern Name	Inputs (choice, n)	Expected Output
1	Right-angled triangle	choice = 1, n = 4	* ** *** ****
2	Row-number triangle	choice = 2, n = 4	1 2 2 3 3 3 4 4 4 4
3	Natural-number pattern	choice = 3, n = 4	1 1 2 1 2 3 1 2 3 4
4	Consecutive-number triangle	choice = 4, n = 4	1 2 3 4 5 6 7 8 9 10
5	Alphabet pattern	choice = 5, n = 4	A A B A B C A B C D

3. C PROGRAM SOURCE CODE

```
#include <stdio.h>

// 1. Increasing right-angled triangle
void pattern1(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
}
```

```

    }
}

// 2. Increasing row-number triangle
void pattern2(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d ", i);
        }
        printf("
");
    }
}

// 3. Increasing natural-number pattern
void pattern3(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d ", j);
        }
        printf("
");
    }
}

// 4. Consecutive-number triangle
void pattern4(int n) {
    int count = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%d ", count);
            count++;
        }
        printf("
");
    }
}

// 5. Increasing alphabet pattern
void pattern5(int n) {
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= i; j++) {
            printf("%c ", 'A' + j - 1);
        }
        printf("
");
    }
}

int main() {
    int choice, n;

    printf("--- MENU ---
");
    printf("1. Increasing right-angled triangle
");
    printf("2. Increasing row-number triangle
");
    printf("3. Increasing natural-number pattern
");
    printf("4. Consecutive-number triangle
");
    printf("5. Increasing alphabet pattern
");
    printf("Enter your choice (1-5): ");
    scanf("%d", &choice);

    if (choice >= 1 && choice <= 5) {

```

```
printf("Enter the number of rows (n): ");
scanf("%d", &n);

switch (choice) {
    case 1:
        pattern1(n);
        break;
    case 2:
        pattern2(n);
        break;
    case 3:
        pattern3(n);
        break;
    case 4:
        pattern4(n);
        break;
    case 5:
        pattern5(n);
        break;
}
} else {
    printf("Invalid choice!");
}

return 0;
}
```